MIL-S-10999C(AT)
20 September 1967
SUPERSEDING
MIL-S-10999B(Ord)
16 April 1957

MILITARY SPECIFICATION

SEAT, INDIVIDUAL EQUIPMENT OPERATOR: AND
SEAT, PERSONNEL: FOR COMBAT VEHICLES

1. SCOPE

1.1 This specification covers crew-personnel seats for use in combat vehicles.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-00386	-	Plastic Material, Cellular, Urethane.
NN-P-530	-	Plywood, Flat Panel.
TT-W-570	-	Wood-Preservative, Pentachlorophenal, Solid.
TT-W-571	-	Wood Preservation: Treating Practices.
CCC-A-700	-	Artificial Leather, Cloth, Coated Vinyl Resin, (Upholstery).
CCC-C-419	-	Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.
CCC-C-428	-	Cloth, Duck, Cotton: Fire, Water, Weather and Mildew Resistant.
CCC-C-429	-	Cloth, Osnaburg, Cotton.
CCC-T-191	-	Textile Test Method.

Military

MIL-P-116	-	Preservation, Methods of.
MIL-C-3133	-	Cellular Elastomeric Materials, Fabricated Parts.
MIL-W-3912	-	Wood Parts, Fabricated: for Transport Vehicle Bodies and Similar Applications.
MIL-H-10368	-	Hair, Bound: for Padding on Military Vehicles.
MIL-S-13518	-	Sealer, Surface, Wood Preservative.
MIL-F-13927	-	Fungus Resistance Tests, Automotive Components.

FSC-2540

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:52Z
Check the source to verify that this is the current version before use.

STANDARDS
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-670 - Classification System and Tests for Cellular Elastomeric Materials.

DRAWING
Army
7527188 - Fabrication and Treatment of Thread for Fabric Items.

(Copies of specifications, standards, drawings, and publications required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. The supplier shall furnish a seat within the time frame specified in the contract or purchase order to prove, prior to starting production, that his production methods will produce seats that comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the preproduction model during production shall be subject to the approval of the contracting officer. Approval of the preproduction model by the activity concerned shall not relieve the supplier of his obligation to furnish seats conforming to this specification.

3.2 Materials. Materials used in the fabrication of seats shall be as specified herein or in accordance with applicable specifications or drawings (see 6.2).

3.2.1 Toxic materials. Seat assemblies shall not contain materials which will cause discomfort or ill effects to personnel from contact with the seats.

3.2.2 Solid wood. Wood used in seat construction shall be one of the following species: white ash, beech, yellow birch, oak, hard elm, hickory, hard pecan or southern pine. Wood shall meet the applicable requirements of MIL-W-3912 and shall be surfaced to the correct cross-sectional dimension before pressure treatment. Treatment shall be to refusal with pentachlorophenol conforming to TT-W-570 dissolved in a completely volatile solvent, in accordance with TT-W-571. The treated wood shall be paintable.

3.2.3 Plywood. When specified (see 6.2), plywood shall be Group B, exterior type, Grade B-B or A-B (Species - Group 1 of U S Product Standard PS-1) conforming to NN-P-530, of a thickness as specified on the applicable drawing. Plywood shall be treated with material conforming to MIL-S-13518 and need not be subjected to vacuum treatment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:52Z
Check the source to verify that this is the current version before use.

3.2.4 Padding filler. Padding filler material shall consist of bound hair, cotton osnaburg or cellular elastomeric material in accordance with 3.2.4.1, 3.2.4.2 or 3.2.4.3.

3.2.4.1 Bound hair. Bound hair padding shall conform to class 2 of MIL-H-10368.

3.2.4.2 Cotton osnaburg. The osnaburg cotton cloth shall be of the unbleached type, class 2, and shall conform to CCC-C-429.

3.2.4.3 Cellular elastomer. Cellular elastomer material shall conform to grade RU 20 of MIL-STD-670 or polyurethane foam, type II, class 3 of L-P-00386.

3.2.5 Covering. Covering shall be artifical leather or canvas duck in accordance with 3.2.5.1 or 3.2.5.2.

3.2.5.1 Artificial leather. Artificial leather, of olive drab color, shall conform to class 4, treatment a, b, and c of CCC-A-700.

3.2.5.2 Canvas duck. Canvas duck shall conform to type I, number 8 of CCC-C-419 and shall be treated in accordance with type 1, class 2 of CCC-C-428.

3.2.6 Thread. Thread shall conform to number 2 of Drawing 7527188.

3.3 Construction and design.

3.3.1 Assembly. Each seat shall be furnished as a complete assembly in accordance with applicable drawings (see 6.2). Each seat shall include at least a supporting frame, a bottom pan, a back-rest pad, cushion upholstery on the back-rest and cushion, and a seat position control arrangement. The assembly shall be free of projecting points, sharp edges, or bulges which could cause injury to personnel or damage to material.

3.3.1.1 Seat frame. The seat assembly main-frame shall support other components of the seat. Supporting members shall be of single or multiple construction. Means shall be incorporated for securely fastening the seat assembly in the intended position. When specified (see 6.2), design of the frame shall incorporate a mechanism for cushioning the seat against vibration or shock incident to vehicle operation over rough terrain. The mechanism shall be easily accessible for servicing, but shall be enclosed to prevent injury to personnel. When incorporated in the seat mechanism, or used as part of the position control arrangement, springs shall be of steel spring wire, or of such design that any forces encountered during operation will be distributed throughout the entire length of the springs.

3.3.1.2 Seat bottom. The bottom pan shall be arranged to hold and retain either a fixed or removable cushion as specified, and shall be provided with openings for draining any water that may accumulate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:52Z
Check the source to verify that this is the current version before use.

3.3.1.3 Fastening devices. Fastening devices shall provide positive means for attaching seat components and shall secure the seat assembly to the vehicle or adjusting device.

3.3.1.4 Adjusting devices. Seat position adjusting devices shall be positive and quickly and easily operated without the use of tools. Adjusting devices shall prevent release or altering of position due to impact or accidental contact with hand or foot. Means shall be provided to securely fasten the adjusting device to the seat assembly and to the vehicle.

3.3.1.5 Back-rest pads. Padding for back-rest pads shall consist of firm or medium density bound hair or cellular elastomer.

3.3.1.6 Full pad design. Full pad cushions shall be padded with heavy density bound hair, overlaid with a cellular elastomer topper, and covered on top, front and side risers with covering applied so that inner surfaces will not be exposed.

3.3.1.7 Pad-over-spring design. Pad-over-spring cushions shall be padded with bound hair, cellular elastomer or combination thereof and shall incorporate upholstery springs covered on the upper surface with cotton osnaburg, and upholstered with a covering in the manner specified in 3.3.1.6. Springs shall be assembled in clusters of two or more units attached to one another to insure long service life and a minimum of bottoming and lateral displacement under load. The bottoms of the springs shall be attached to the seat structure, except that when cushions are of the demountable type, the bottoms of the springs shall be covered with and secured to a layer of cotton osnaburg.

3.3.1.8 Ventilation. Grommet breathers installed in cushion sides, and drilled holes in the seat bottom, of a size and quantity specified on applicable drawings, shall be incorporated in all seats.

3.3.1.9 Dimensions. Dimensions of the seat assembly, including pads, backrests and cushions, shall conform to applicable drawings.

3.3.2 Color. Color of seat main-frame, back-rests, pads, and cushions shall be as specified by the procuring activity (see 6.2).

3.4 Performance.

3.4.1 High temperature resistance. When tested as specified in 4.7.1, the seat shall show no evidence of damage or deterioration. Artificial leather or other approved covering shall not become tacky or sticky and shall not emit odors or stain clothing.

3.4.2 Low temperature resistance. When tested as specified in 4.7.2, seats and components shall show no evidence of damage or deterioration. Artificial leather, or other approved material, shall be free of cracks and checks.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:52Z
Check the source to verify that this is the current version before use.

3.4.3 Load capacity.

3.4.3.1 Complete seat. Each seat shall show no evidence of damage, or no more than 5 percent change in original dimensions of any component after being tested in accordance with 4.7.3.

3.4.3.2 Cushion or cushion pad only. After testing in accordance with 4.7.3, the removable cushion or cushion pad shall retain 90 percent of its original shape and dimensions.

3.4.4 Compression set. When tested as specified in 4.7.4, the cushion or cushion pad shall retain the following percentage of its original height:

3.4.4.1 At plus 170°F. Compression set at plus 170°F shall be not less than 80 percent.

3.4.4.2 At minus 65°F. Compression set at minus 65°F shall be not less than 75 percent.

3.4.5 Resistance to impact. When tested as specified in 4.7.5, the components of the seat assembly shall withstand the amount and number of impacts (see table I) without evidence of broken springs, flattened padding, torn upholstery, split wood, deformed metal bottom pan, or loosening or shifting of adjusting device.

Table I. Impact requirements for seat components

	Kind of assembly	Support	Impact, foot-pounds ±2	Number of impacts
Seat	Without cushion or pad	Single	35	2,000
		Multiple	57	2,000
	With cushion or pad	Single	57	2,500
		Multiple	70	2,500
Seat back or back rest	Without cushion or pad	Single	35	1,000
		Multiple	35	1,000
	With cushion or pad	Single	35	1,500
		Multiple	35	1,500
Removable cushion or pad			70	5,000

3.4.6 Flame resistance. When tested as specified in 4.7.7, the material shall not flame more than 2 seconds or glow more than 15 seconds after the flame has been withdrawn.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:52Z
Check the source to verify that this is the current version before use.

3.4.7 Fungus resistance. Upholstery material shall be treated to prevent damage by the growth of fungus. Materials, after being tested as specified in 4.7.6, shall retain a tensile strength of not less than 90 percent of its original tensile value.

3.5 Workmanship. Workmanship shall be such as to produce finished seats that are free from cracks or porosity in the outer coating. The sides shall be as free of seams and laps as practicable. Surfaces of the outer face and sides shall be smooth and free of bumps, blisters, or abrasions. The supporting frame shall be free from sharp edges or corners.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 Preproduction sample. One complete sample (see 3.1) shall be furnished for preproduction testing as specified in 4.6.1. The preproduction sample shall be examined for defects listed in table II.

4.3 Failure. Failure of a preproduction sample to satisfy any of the requirements specified herein will be cause for refusal by the Government to conduct additional inspection until satisfactory corrective measures have been taken by the contractor.

4.4 Quality conformance inspection.

4.4.1 Lot formation. A lot shall consist of all seats of one kind and size from an identifiable production period from one manufacturer, submitted at one time for quality conformance inspection.

4.4.2 Sampling for examination. For visual, dimensional and primary functional inspection, a representative sample shall be selected from each lot in accordance with inspection level II of MIL-STD-105.

4.4.3 Sampling for quality conformance.

4.4.3.1 Normal sampling. From each lot which has passed inspection specified in 4.4.2, a representative sample shall be selected for quality conformance tests specified in 4.6.2 in accordance with inspection level S-3 of MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:52Z
Check the source to verify that this is the current version before use.

4.4.3.2 Reduced sampling. Reduced sampling may be initiated provided the conditions for reduced inspection procedure R1 of MIL-STD-105 have been met. The inspection level for reduced testing shall comply with inspection level S-2 of MIL-STD-105.

4.4.4 Sampling for control tests. Sampling for control tests shall be one complete seat assembly from each 100 production assemblies, but not more than 2 sets per month.

4.5 Acceptable quality level. Samples selected in accordance with 4.4.2 shall be examined for conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.5.1 Classification of defects. For examination purposes, defects shall be as classified in table II.

Table II. Classification of defects

Categories	Defects
Critical	None.
Major:	
101	Construction and design (see 3.3)
102	Assembly (see 3.3.1)
103	Seat frame (see 3.3.1.1)
104	Seat bottom (see 3.3.1.2)
105	Fastening devices (see 3.3.1.3)
106	Adjusting devices (see 3.3.1.4)
107	Back-rest pads (see 3.3.1.5)
108	Full pad design (see 3.3.1.6)
109	Pad-over-spring design (see 3.3.1.7)
110	Ventilation (see 3.3.1.8)
111	Dimensions (see 3.3.1.9)
Minor:	
201	Color (see 3.3.3)
202	Workmanship (see 3.5)

4.6 Tests.

4.6.1 Preproduction tests. The preproduction sample (see 3.1 and 4.2) shall be subjected to inspection and all tests specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:52Z
Check the source to verify that this is the current version before use.

4.6.1.1 Failure. Failure of any preproduction test specimen to pass inspection or any specified test shall be cause for refusal by the Government to conduct additional tests until it has been proved to the Government activity that the faults revealed have been corrected by the contractor.

4.6.2 Quality conformance. Seats selected for quality conformance tests (see 4.4.3) shall be subjected to tests specified in 4.7.3, 4.7.5, and 4.7.7, using AQL of 6.5 defective.

4.6.3 Control tests. Samples selected for control test (see 4.4.4) shall be subjected to tests specified in 4.7.1, 4.7.2, 4.7.4 and 4.7.6.

4.6.3.1 Failure. If an item fails to pass a control test specified herein, two additional specimens shall be tested. In the event of failure during retest, the Government inspector shall stop acceptance of subsequent lots until evidence has been provided by the manufacturer that corrective action has been taken.

4.7 Test procedures.

4.7.1 High temperature test. When tested as specified herein, the seats shall conform to 3.4.1.

4.7.1.1 Apparatus. Apparatus shall include a chamber capable of maintaining an ambient air temperature of 125°F.

4.7.1.2 Procedure. A seat specimen shall be placed in the test chamber and subjected to an ambient air temperature of 125°F for a period of thirty days. It shall then be examined for conformance to 3.4.1.

4.7.2 Low temperature test. When tested as specified herein, the seats shall conform to 3.4.2.

4.7.2.1 Apparatus. A chamber capable of maintaining an ambient air temperature of -65°F.

4.7.2.2 Procedure. A seat specimen shall be placed in the test chamber and subjected to an ambient air temperature of -65°F for a period of thirty days. It shall then be examined for conformance to 3.4.2.

4.7.3 Load capacity test. A sample seat shall be tested as specified herein to determine conformance to 3.4.3.

4.7.3.1 Specimen. One seat shall be furnished for this test.

4.7.3.2 Apparatus. The following testing apparatus shall be provided, and used as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:52Z
Check the source to verify that this is the current version before use.

4.7.3.2.1 Loading form. This shall be a steel plate, form of shape to maintain contact with the major area of the seat load-carrying member, and to present a relatively flat surface during the test.

4.7.3.2.2 Test weight. This shall consist of a 350-pound weight with a base diameter of 11 ± 1 inches, and may be a cylindrical bag filled with shot (or similar loose material) or of solid material with a circular flat base.

4.7.3.3 Procedure.

4.7.3.3.1 Stripped seats. The test specimen shall be mounted, less cushion or back, in a manner similar to that in which it will be installed in the intended vehicle and the loading form placed in position on the seat bottom. It shall be adjusted during the test in the top or bottom vertical position, and in the middle horizontal position (when provided). The test weight shall be dropped 5 times from a height of 3-1/2 feet on the seat load-carrying member (or on the cushion if of fixed design). At the end of this test the seat shall be examined for conformance to 3.4.3.1.

4.7.3.3.2 Removable cushion or pad. The cushion or cushion pad (not tested with the seat assembly) shall be placed top side up, or horizontal, on a reasonably level foundation, and tested in accordance with 4.7.3.3.1 except that loading form will not be used, and the weight shall be placed and left for 24 hours on the seat. The specimen shall then be removed and examined for conformance to 3.4.3.2.

4.7.3.3.3 Seat back or back rest. Seat backs or back rests, incorporated in the seat assembly, shall be subjected to the load-capacity test by mounting in a relatively horizontal position with the load carrying surface facing upward and in accordance with the test procedure of 4.7.3.3.2. Pads of removable design and irregular shape shall be tested in a like manner except that parts shall be held on an appropriate wooden form, mounted on a substantial horizontal base. After testing, they shall be examined for conformance to 3.4.3.2.

4.7.4 Compression set test. After testing as specified herein, the cushion or cushion pad shall conform to all the requirements specified in 3.4.4.

4.7.4.1 Compression set at ∠170°F. The cushion or pad of the seat assembly shall be subjected to an ambient air temperature of ∠170°F for 48 hours. After 47 hours the thickness of the cushion or pad shall be measured and the measurement recorded. The test shall continue for the final hour with the test weight (see 4.7.3.2.2) imposed upon the upper surface of the cushion. With the temperature still held at ∠170°F, the weight shall be removed and after 5 minutes, the thickness shall again be determined. Compression set shall be calculated as the percentage of final to original thickness (see 3.4.4) .

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:52Z
Check the source to verify that this is the current version before use.

4.7.4.2 Compression set at -65°F. The cushion or pad of the seat assembly shall be subjected to an ambient air temperature not in excess of -65°F for 24 hours. After 23 hours the thickness of the cushion or pad shall be measured and the measurement recorded. The test shall continue for the final hour with the test weight (see 4.7.3.2.2) imposed upon the upper surface of the cushion. With the temperature still held at -65°F, the weight shall be removed and after 5 minutes the thickness shall again be determined. The compression set shall be calculated as the percentage of final to original thickness (see 3.4.4).

4.7.5 Resistance to impact test. When tested as specified in table I, the seat components shall conform to 3.4.5.

4.7.5.1 Test specimen. One seat shall be furnished for this test.

4.7.5.2 Apparatus. The apparatus shall consist of a test striker and a device for delivering free-falling impacts at the rate of 20 ± 2 blows per minute. The striker shall be of the shape and dimensions as specified in 4.7.3.2.2, and of a weight which will produce the foot-pound impacts shown in table I.

4.7.5.3 Procedure. The seat cushion or pad shall be mounted and positioned as specified in 4.7.3.3. The cushion or pad, if of removable design, shall be secured face upward on a substantial and reasonably level foundation. It shall then be subjected, at a temperature of 80° ± 20°F, to impacts in number and amount shown in table I, delivered in a direction generally perpendicular to the load surface, and at the rate of approximately 20 impacts per minute. Each component shall be tested first with the striker arranged so that its center will impact the center of the component load surface, and then with the weight arranged so that its center will impact the load surface at a point approximately 2 inches from its front edge near its center line. At the conclusion of these tests the components shall be examined for conformance to 3.4.5.

4.7.6 Fungus resistance. Artificial leather shall be tested for fungus resistance in accordance with MIL-F-13927. Tensile strength of these materials, before and after exposure to the fungus resistance test, shall conform to 3.4.7.

4.7.7 Flame resistance. The flame resistance test shall be conducted in accordance with test method 5902 of CCC-T-191 and shall conform to 3.4.6.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Seat assemblies shall be preserved, packaged, packed, and marked in accordance with the applicable packaging standard or packaging data sheet, as specified by the procuring activity (see 6.2).

6. NOTES

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:52Z
Check the source to verify that this is the current version before use.

6.1 Intended use. The seats covered by this specification are intended for use in wheeled and tracked combat vehicles, to provide seating accommodations for the vehicle operator or for the crew in personnel compartments.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Title, number and date of applicable drawings.
(c) Time schedule for submission of preproduction sample (see 3.1).
(d) Materials (see 3.2).
(e) If plywood other than specified is required (see 3.2.3).
(f) If mechanism for cushioning vibration and shock as specified is required (see 3.3.1.1).
(g) Color (see 3.3.2).
(h) Selection of applicable packaging standard or packaging data sheet (see 5.1).

Custodian:
Army - AT

User activity:
Army - SM

Preparing activity:
Army - AT

Project No. 2540-A118

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:52Z
Check the source to verify that this is the current version before use.